

# Astra, Opus and the revolved mesher

## Three solid profiles

Prepared by Ryan | 8 October 2026

Independent LLM-authored block layouts and deterministic CAD-to-mesh results for three bore-free solids of revolution. The examples vary stepped shoulders, chamfered tips, rounded ends and an exact circular fillet.

| Part | Approach        | Status  | Hexes | Min SJ / doubled | Checks | Time    | Usage* |
|------|-----------------|---------|-------|------------------|--------|---------|--------|
| 01   | GPT 6 Astra     | Pass    | 2,898 | 0.6917 / 0.6967  | 1      | 2.22 m  | \$1.01 |
| 01   | Opus 5.5        | Pass    | 2,464 | 0.6946 / 0.6996  | 1      | 3.02 m  | \$0.73 |
| 01   | Revolved mesher | Pass    | 8,064 | 0.6924 / 0.6935  | 1      | 0.024 s | No API |
| 02   | GPT 6 Astra     | Pass    | 4,500 | 0.6964 / 0.7006  | 1      | 2.28 m  | \$1.02 |
| 02   | Opus 5.5        | Pass    | 3,584 | 0.6946 / 0.6996  | 1      | 4.03 m  | \$0.93 |
| 02   | Revolved mesher | Pass    | 6,180 | 0.6855 / 0.6860  | 1      | 0.043 s | No API |
| 03   | GPT 6 Astra     | Pass    | 1,824 | 0.5774 / 0.5774  | 1      | 2.22 m  | \$0.93 |
| 03   | Opus 5.5        | Pass    | 1,740 | 0.6428 / 0.6428  | 1      | 6.55 m  | \$1.34 |
| 03   | Revolved mesher | Refused | -     | - / -            | 0      | 0.002 s | No API |

\*GPT dollars are Standard API-equivalent estimates from cumulative native tokens; Claude dollars are CLI-reported usage. Neither is a subscription invoice. Deterministic time covers compilation; model time covers the author session. Astra uses medium reasoning, Opus high.

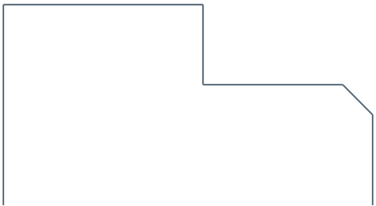
Explicitly synthetic showcase, not a population benchmark. The original CAD and native geometry checks are shared within this experiment. The filleted case records the current automatic-route refusal honestly. No prior or other-model layouts supplied to authors; no expert grades assigned.

# Stepped shaft with chamfered nose

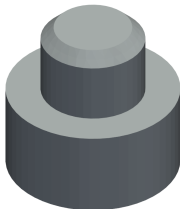
## Original CAD

Full solid of revolution.  
No holes or internal cavities.  
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



### Views and quality

Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

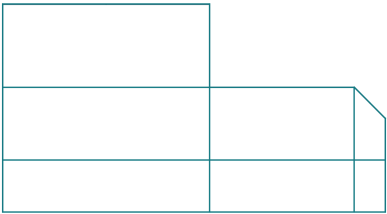
## GPT 6 Astra

### Geometry pass

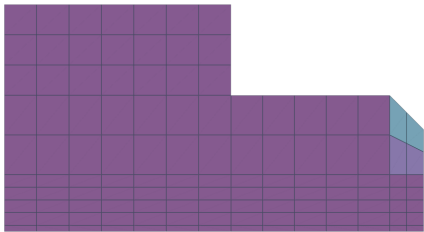
9 seed parts | 2898 hexes  
SJ 0.6917 / 2x 0.6967  
1 full checks | 2.22 min  
Usage \$1.01

Standard API equivalent

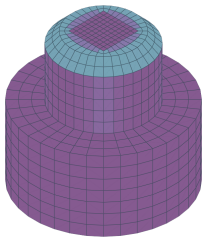
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



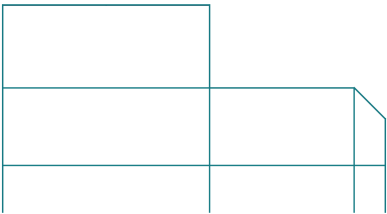
## Opus 5.5

### Geometry pass

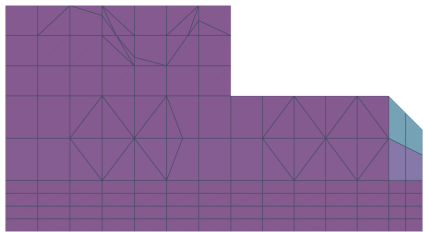
5 seed parts | 2464 hexes  
SJ 0.6946 / 2x 0.6996  
1 full checks | 3.02 min  
Usage \$0.73

Claude CLI usage

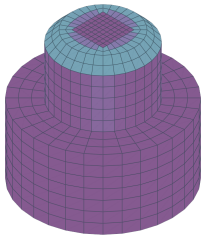
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



## Revolved mesher

### Geometry pass

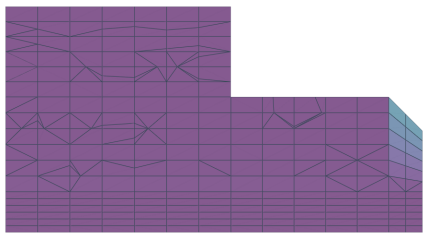
5 seed parts | 8064 hexes  
SJ 0.6924 / 2x 0.6935  
1 full checks | 0.024 s  
Usage no API

local deterministic computation

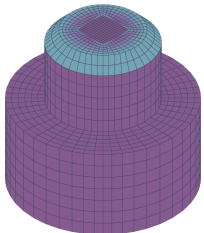
REALIZED SEED PARTS



MESH SECTION / SJ



3D MESH / SJ



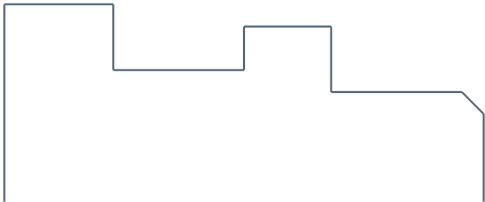
SJ 0.2-1; larger is better. Seed parts may contain several blocks.

# Multi-step shaft with chamfered tip

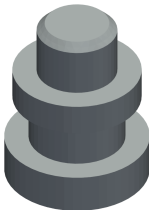
## Original CAD

Full solid of revolution.  
No holes or internal cavities.  
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



**Views and quality**  
Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

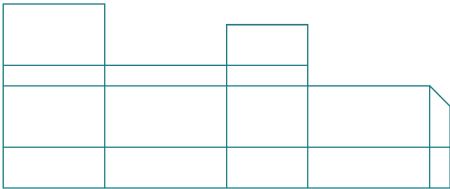
## GPT 6 Astra

### Geometry pass

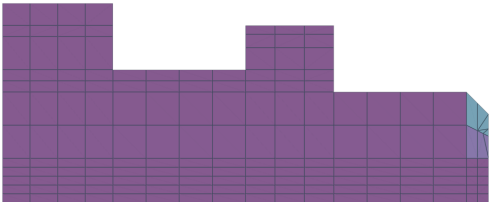
17 seed parts | 4500 hexes  
SJ 0.6964 / 2x 0.7006  
1 full checks | 2.28 min  
Usage \$1.02

Standard API equivalent

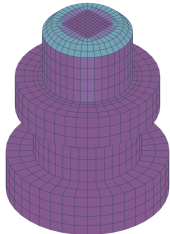
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



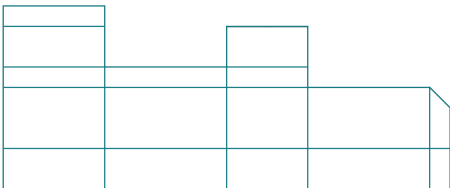
## Opus 5.5

### Geometry pass

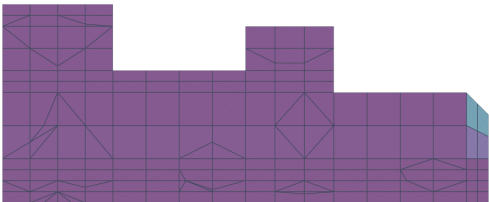
1 seed parts | 3584 hexes  
SJ 0.6946 / 2x 0.6996  
1 full checks | 4.03 min  
Usage \$0.93

Claude CLI usage

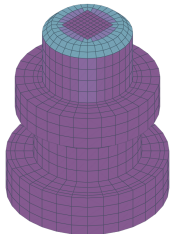
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



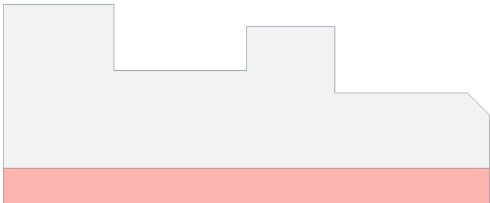
## Revolved mesher

### Geometry pass

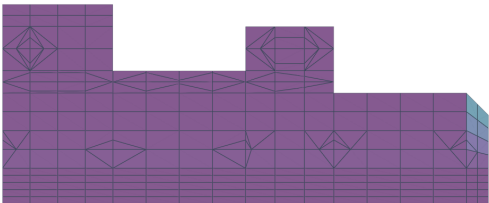
5 seed parts | 6180 hexes  
SJ 0.6855 / 2x 0.6860  
1 full checks | 0.043 s  
Usage no API

local deterministic computation

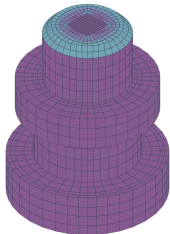
REALIZED SEED PARTS



MESH SECTION / SJ



3D MESH / SJ



SJ 0.2-1; larger is better. Seed parts may contain several blocks.

# Filletted tapered body with rounded ends

## Original CAD

Full solid of revolution.  
No holes or internal cavities.  
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



### Views and quality

Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

Fillet detail: Astra represents the tiny fillet with 48/192 boundary quads; Opus spans it (0 attributed quads). Both pass native checks; neither has an eligible sharp edge in this case.

## GPT 6 Astra

### Geometry pass

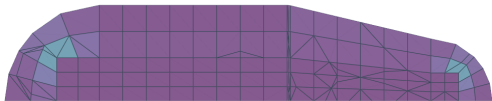
27 seed parts | 1824 hexes  
SJ 0.5774 / 2x 0.5774  
1 full checks | 2.22 min  
Usage \$0.93

Standard API equivalent

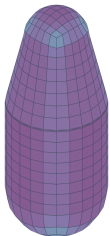
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



## Opus 5.5

### Geometry pass

1 seed parts | 1740 hexes  
SJ 0.6428 / 2x 0.6428  
1 full checks | 6.55 min  
Usage \$1.34

Claude CLI usage

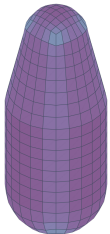
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



## Revolved mesher

### Refused

- seed parts | - hexes  
SJ - / 2x -  
0 full checks | 0.002 s  
Usage no API

local deterministic computation

No layout or mesh emitted.

The automatic route refuses a shallow surface junction. Its two-surface projection construction cannot guarantee convergence here. The CAD and checks were not changed.



SJ 0.2-1; larger is better. Seed parts may contain several blocks.